

Physics Midterm #1

Unit 01. a) $q = 1 \mu\text{C} = 1 \times 10^{-6} \text{C}$ $x = 8 \text{ cm}$ $+q = 3 \text{ cm}$ $-2q = 8 \text{ cm}$ $+q = 11 \text{ cm}$

$$F = K \frac{q_1 q_2}{r^2}$$

$$r = 8 - 3 = 5 \text{ cm} \rightarrow .05 \text{ m}$$

$$F = (9 \times 10^9) \frac{(1 \times 10^{-6} \text{C})(-2 \times 10^{-6} \text{C})}{(.05)^2}$$

$$(9 \times 10^9) \frac{(2 \times 10^{-6})}{.0025} = (9 \times 10^9) \times .0008 = (8 \times 10^{-4}) = 7.2 \text{ N left}$$

b) $x = 11 \text{ cm}$

$$E = K \frac{q}{r^2}$$

$$-2q = 1 \text{ cm}$$

$$-2q = 1 \text{ cm}$$

$$r = .1 \text{ m}$$

$$F = (9 \times 10^9) \frac{2 \times 10^{-6}}{.1^2}$$

$$= 1800 \text{ N/C}$$

left

$$+q = 5 \text{ cm}$$

$$+q = 5 \text{ cm}$$

$$r = .06 \text{ m}$$

$$E = (9 \times 10^9) \frac{1 \times 10^{-6}}{.06^2}$$

$$= 2500 \text{ N/C}$$

right

$$+3q = 8 \text{ cm}$$

$$+3q = 8 \text{ cm}$$

$$r = .03 \text{ m}$$

$$E = (9 \times 10^9) \frac{(3 \times 10^{-6})}{.03^2}$$

$$= 30000 \text{ N/C}$$

right

$$-q = 4 \text{ cm}$$

$$-q = 4 \text{ cm}$$

$$r = .03 \text{ m}$$

$$E = (9 \times 10^9) \frac{(1 \times 10^{-6})}{.03^2}$$

$$= 10000 \text{ N/C}$$

left

a) continued

$$+q = 11 \text{ cm}$$

$$r = 11 - 8 = .03 \text{ m}$$

$$F = (9 \times 10^9) \frac{(1 \times 10^{-6})(2 \times 10^{-6})}{.03^2}$$

$$(9 \times 10^9) (2.22 \times 10^{-9})$$

$$= 20 \text{ N right}$$

$$\text{Net} = 20 \text{ N} - 7.2 \text{ N} = 12.8 \text{ N}$$

right

left

right

2. a)



$$q_a = q_b = +7.5 \mu\text{C}$$

$$+7.5 + (-7.5) = F_{\text{net}} = 0 \text{ N}$$

$$q_c = q_d = -7.5 \mu\text{C}$$

b) $q = 2 \mu\text{C}$

$$r = 10 \text{ cm} = .1 \text{ m}$$

$$K = 9 \times 10^9$$

$$d = \frac{r}{\sqrt{2}} = \frac{.1}{\sqrt{2}} = .071 \text{ m}$$

$$F = (9 \times 10^9) \frac{(2 \times 10^{-6})(7.5 \times 10^{-6})}{.071^2} = (9 \times 10^9) (3 \times 10^{-9}) = 27 \text{ N}$$

3. 40 kV

$$qV = \frac{1}{2}mv^2$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$V = 4 \times 10^4$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$v = \sqrt{\frac{2qV}{m}}$$

$$v = \sqrt{\frac{2(1.6 \times 10^{-19})(4 \times 10^4)}{9.1 \times 10^{-31}}} = 1.19 \times 10^8 \text{ m/s}$$

4. $4 \text{ cm} = 7.5 \times 10^4 \text{ V/m}$

a) $\Delta V = Ed$

$$7.5 \times 10^4 (.04 \text{ m})$$

$$= 3 \times 10^3 \text{ V}$$

b) $.01 \text{ m lower}$

$$V = 7.5 \times 10^4 (.01 \text{ m})$$

$$= 750 \text{ V}$$

c) $E = \frac{\Delta V}{d}$

$$= \frac{80 \text{ mV}}{9 \text{ nm}} = \frac{8 \times 10^{-2}}{9 \times 10^{-9}}$$

$$8.89 \times 10^6 \text{ V/m}$$

5. 32 keV $2 \text{ cm} = .02 \text{ m}$
 $E = 32 \times 10^3 \times 1.602 \times 10^{-19} \text{ J}$
 $q = 1 \times 1.602 \times 10^{-19} \times 2 = 3.204 \times 10^{-19} \text{ C}$
 $\Delta V = \frac{E}{q} = \frac{32 \times 10^3 \times 1.602 \times 10^{-19}}{3.204 \times 10^{-19}} = \frac{16000}{.02} = 8 \times 10^5 \text{ V/m}$

Unit 1 1. $3 \mu\text{C}$ @ 120 V
 $C = \frac{Q}{V} = \frac{3 \times 10^{-6} \text{ C}}{120 \text{ V}} = 2.5 \times 10^{-8} \text{ F}$

2. a) $10 \mu\text{C}$ $9 \times 10^3 \text{ V}$
 $E = \frac{1}{2} CV^2 = \frac{1}{2} (10 \times 10^{-6}) (9 \times 10^3) = \frac{1}{2} (10 \times 10^{-6}) (8.1 \times 10^7)$
 $4.05 \times 10^2 \text{ J}$
 b) $8 \mu\text{F}$ 40 J
 $\rightarrow V = \sqrt{\frac{2E}{C}} = \sqrt{\frac{2(40)}{8 \times 10^{-6}}} = \sqrt{\frac{80}{8 \times 10^{-6}}} = \sqrt{10^7} = 3162.3 \text{ V}$

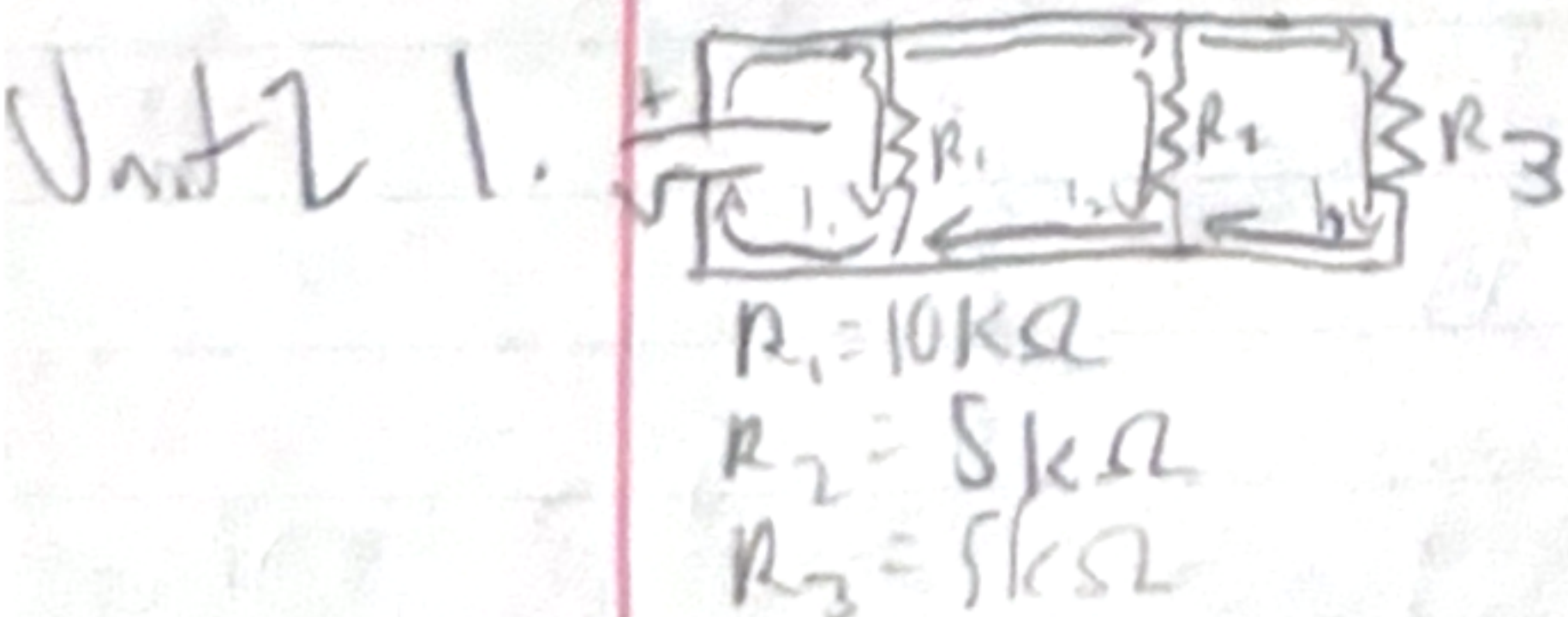
3. a) $E = 405 \text{ J}$ $V = 9 \times 10^3$ $C_{\text{tot}} = C_1 + C_2 + C_3 + C_4 = 4 \text{ C}$
 $405 = \frac{1}{2} (4 \text{ C}) (9 \times 10^3)^2$
 $= 2 \text{ C} (8.1 \times 10^7)$
 $= 1.62 \times 10^8 \text{ C}$
 $C = \frac{405}{1.62 \times 10^8} = 2.5 \times 10^{-8} \text{ F} = 2.5 \mu\text{F}$

b) There's more charge when they're parallel

4. $1 \text{ k}\Omega$ resistor 3 V battery

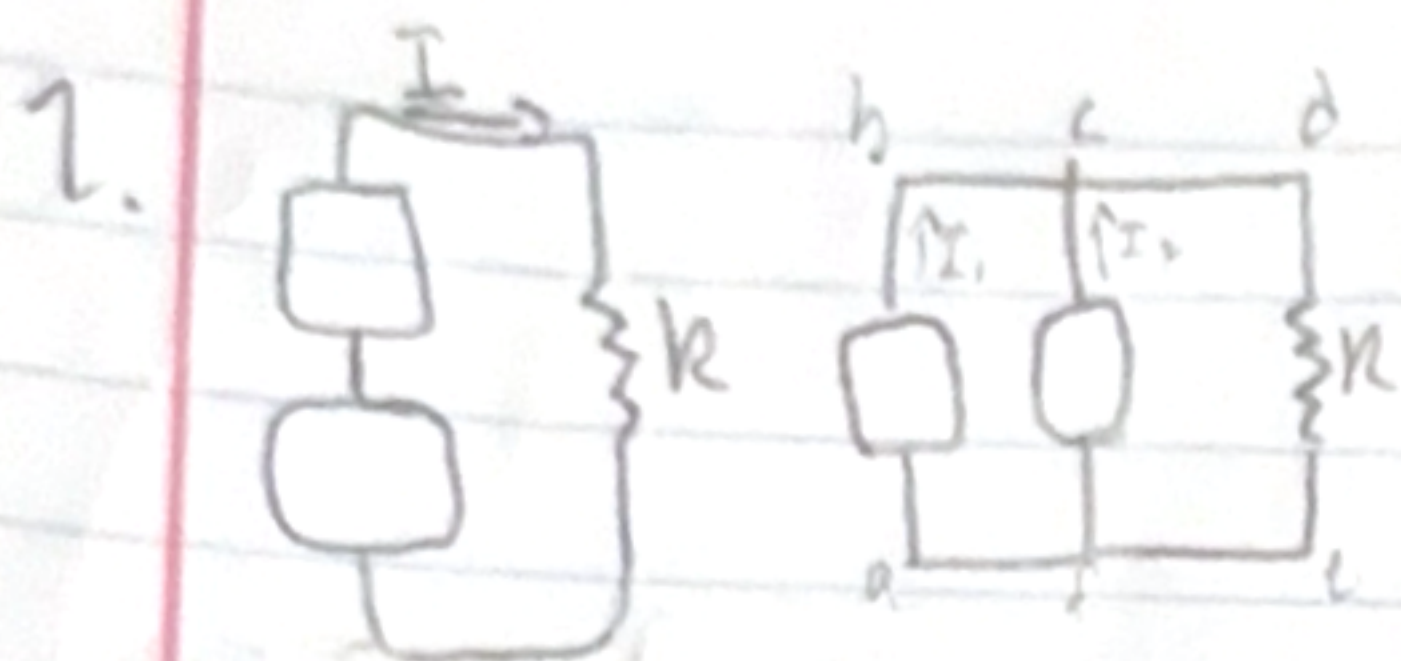
a) $R_{\text{LED}} = 3 \Omega$
 $R_{\text{res}} = 1 \text{ k}\Omega \rightarrow 1000 \Omega$
 $V_{\text{Battery}} = 3 \text{ V}$
 $R_{\text{tot}} = 1003 \Omega$
 $I = \frac{V}{R_{\text{tot}}} = \frac{3 \text{ V}}{1003 \Omega} \approx .003 \text{ A} \approx 3 \text{ mA}$

b) $P = (.003 \text{ A})^2 \cdot 3 \Omega \rightarrow 9.94 \times 10^{-6} \text{ W} \times 3 \Omega \rightarrow 2.98 \times 10^{-5} \text{ W}$
 2.7 mW



c) $I_1 = \frac{12}{10000 \Omega} = .0012 \text{ A} = 1.2 \text{ mA}$
 $I_2 = \frac{12}{5000 \Omega} = .0024 \text{ A} = 2.4 \text{ mA}$
 $I_3 = \frac{12}{5000 \Omega} = .0024 \text{ A} = 2.4 \text{ mA}$

a) $\frac{1}{R_{\text{tot}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \rightarrow \frac{1}{R_{\text{tot}}} = \frac{1}{10000} + \frac{1}{5000} + \frac{1}{5000} \rightarrow \frac{1}{R_{\text{tot}}} = \frac{3}{10000} + \frac{1}{5000} = \frac{1}{10000} + \frac{2}{5000} = \frac{1}{10000} + \frac{4}{10000} = \frac{5}{10000} \rightarrow \frac{1}{R_{\text{tot}}} = \frac{5}{10000} \rightarrow R_{\text{tot}} = 2000 \Omega$
 b) $R_{\text{tot}} = 2000 \Omega$ $\Delta V = 12 \text{ V}$ $I = \frac{12 \text{ V}}{2000 \Omega} = .006 \text{ A} \rightarrow 6 \text{ mA}$
 5000Ω
 $5 \text{ k}\Omega$



a) $R = 500 \Omega$

$1.5 + 1.5 = 3V$

$I = \frac{V}{R} = \frac{3}{500} = .006 A = 6mA$

$P = I^2 R = (.006^2)(500) = .018W = 18mW$

Series

Parallel

$I = \frac{3V}{500\Omega} = 6mA$

$P = (.006^2) 500 = 18mW$

3. $3 \times 1.58V$

1.5Ω cell

10Ω load resistance

$.02 \Omega$ resistance

$.1 \Omega$ resistance

a) $I = \frac{V}{R} = \frac{(3 \times 1.58V) + 1.5V}{(.02 \Omega) + 10.1 \Omega} = \frac{6.27}{10.12 \Omega} = 617 mA$

b) $P = I^2 R \rightarrow (.617^2) \times 10 \Omega = 3.8W$

c) $I = \sqrt{\frac{P}{R}} = \sqrt{\frac{3.8}{10}} = \sqrt{.38} = .617A$

Unit 3.1.

Case	V direction	B direction	E direction
a	$\hat{i}$ right	$\hat{j}$ up	$-\hat{k}$ into page
b	$\hat{j}$ up	$\hat{k}$ out of page	$\hat{i}$ left
c	$\hat{k}$ out of page	$\hat{i}$ left	$\hat{j}$ down

a) $\hat{i} \times \hat{j} = \hat{k}$

$V \rightarrow$

$B \uparrow$

b) $\hat{j} \times \hat{k} = \hat{i}$

$V \uparrow$

$\hat{k}$ out

$\hat{i} \leftarrow$

c) $\hat{k} \times \hat{i} = \hat{j}$

$\hat{k}$ out

$\hat{i} \leftarrow$

$\hat{j} \downarrow$

2. $V = 4 \times 10^3 m/s$ $B = 1.25T$ $F = 1.4 \times 10^{-16} N$ $q = 1.6 \times 10^{-19} C$

$F = qvB \sin \theta$

$\sin \theta = \frac{F}{qvB} \rightarrow \frac{1.4 \times 10^{-16}}{(1.6 \times 10^{-19})(4 \times 10^3)(1.25)} = \frac{1.4 \times 10^{-16}}{8 \times 10^{-16}} = .175$

$\arcsin .175$

$\approx 10.1^\circ$